

This README is designed to serve as the master context for **Anti-Gravity**. It provides the AI with the necessary logic, technical constraints, and product vision to generate code that is specifically optimized for the Indian financial ecosystem and the Gemini 3 Hackathon.

FinPulse 360: Conversational Expense Intelligence

"The Financial Consciousness Layer for Digital India"

1. Project Overview

FinPulse 360 is an AI-native expense manager designed to solve the "Cryptic Merchant" problem in India. Traditional apps fail because they rely on messy SMS data or delayed bank statements. FinPulse 360 uses a **Hybrid Detection Engine** and **Gemini 3** to capture transaction intent in the "10-second golden window" immediately following a payment.

The Problem

- **Cryptic Merchants:** UPI IDs and bank SMS often show names like "RAJESH_STORES_99" or "91823XXXXX@paytm," making auto-categorization nearly impossible.
- **Friction:** Users hate opening apps to manually fix categories.
- **Security Constraints:** Android's `FLAG_SECURE` blocks screen-reading of many banking apps, and SMS delivery is often unreliable or delayed.

2. Technical Architecture (For AI Implementation)

The app operates on a multi-layered detection stack to ensure 100% reliability without manual entry.

A. The Detection Stack (The "Triggers")

- Still Figuring out the best way...

B. The AI Reasoning Engine (Gemini 3)

- **Gemini 3 Flash:** Handles real-time Hinglish-to-Category extraction.
 - *Input:* "Bought doodh and bread for 150."
 - *Output:* { "category": "Groceries", "amount": 150, "items": ["Milk", "Bread"] }
 - **Gemini 3 Pro:** Conducts the "9 PM Deep Think" session.
 - *Goal:* Reconciles delayed SMS/Account Aggregator data with the user's earlier conversational responses.
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3. Core Features & UX

- **Actionable Notifications:** No app open required. Users reply to a notification with text or voice (via Web Speech API).
- **Multimodal Hinglish Support:** Native understanding of code-switched Hindi/English.
- **Privacy-First:** On-device processing of transaction data; only anonymized text strings are sent to Gemini for categorization.
- **Automated Split-Billing:** AI detects mentions of multiple categories in a single response (e.g., "1500 for petrol and 200 for snacks").

4. Implementation Goals for Anti-Gravity

When generating code or logic, prioritize the following:

1. **Low Latency:** The notification must appear within 5 seconds of a transaction.
2. **Robust Regex:** Account for the varying SMS formats of major Indian banks (HDFC, ICICI, SBI) and UPI apps.
3. **Hinglish Prompt Engineering:** Ensure the System Instruction for Gemini 3 includes a "Category Map" specific to Indian spending habits (e.g., Chai, Kirana, Auto-rickshaw).

5. Development Phases

- **Phase 1:** Build the Android `NotificationListenerService` and `Notiffee` integration for actionable alerts.
- **Phase 2:** Integrate Gemini 3 API for real-time Hinglish parsing.
- **Phase 3:** Implement the "Share Intent" receiver for multimodal receipt analysis.
- **Phase 4:** Develop the 9 PM interactive summary dashboard.

6. Target Context (India-Specific)

- **Currency:** INR (₹).
- **Primary Payment Methods:** UPI (Google Pay, PhonePe, Paytm), IMPS, Credit/Debit Cards.
- **Language:** English + Hinglish (Hindi written in Roman script).

Instruction to AI: "Use this README as the ground truth for all code generation. When writing background services, ensure they are optimized for Android's battery management (Doze Mode) using Foreground Services."